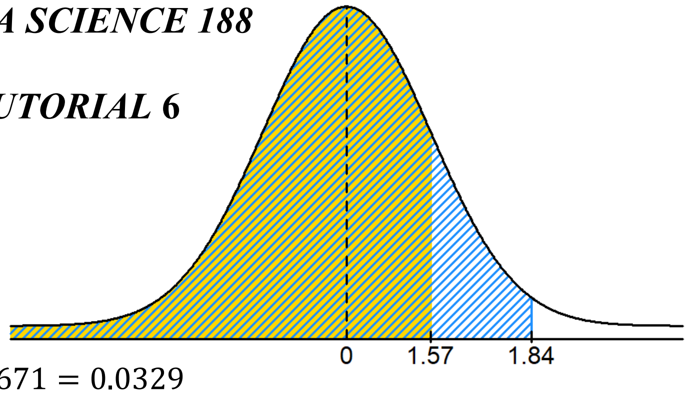
DEPARTEMENT STATISTIEK EN AKTUARIËLE WETENSKAP
DEPARTEMENT OF STATISTICS AND ACTUARIAL SCIENCE

STATISTIEK EN DATAWETENSKAP 188
STATISTICS AND DATA SCIENCE 188

TUTORIAAL 6 / *TUTORIAL 6*

Probleem 1 / *Problem 1*



a) $\mathrm{P}(Z < 1.57) = 0.9418$

b) $\mathrm{P}(Z < 1.84) = 0.9671$

c) $\mathrm{P}(Z > 1.84) = 1 - \mathrm{P}(Z < 1.84) = 1 - 0.9671 = 0.0329$

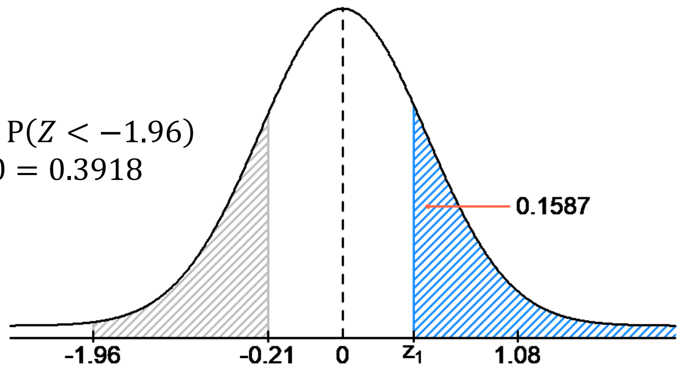
d) $\mathrm{P}(1.57 < Z < 1.84) = \mathrm{P}(Z < 1.84) - \mathrm{P}(Z < 1.57)$
$= 0.9671 - 0.9418 = 0.0253$

e) $\mathrm{P}(Z < 1.57 \; or \; Z > 1.84) = \mathrm{P}(Z < 1.57) + \mathrm{P}(Z > 1.84) - \mathrm{P}(Z < 1.57 \; and \; Z > 1.84)$
$= 0.9418 + 0.0329 + 0 = 0.9747$

Probleem 2 / *Problem 2*

a) $\mathrm{P}(Z > 1.08) = 1 - \mathrm{P}(Z < 1.08) = 1 - 0.8599 = 0.1401$

b) $\mathrm{P}(Z < -0.21) = 0.4168$



c) $\mathrm{P}(-1.96 < Z < -0.21) = \mathrm{P}(Z < -0.21) - \mathrm{P}(Z < -1.96)$
$= 0.4168 - 0.0250 = 0.3918$

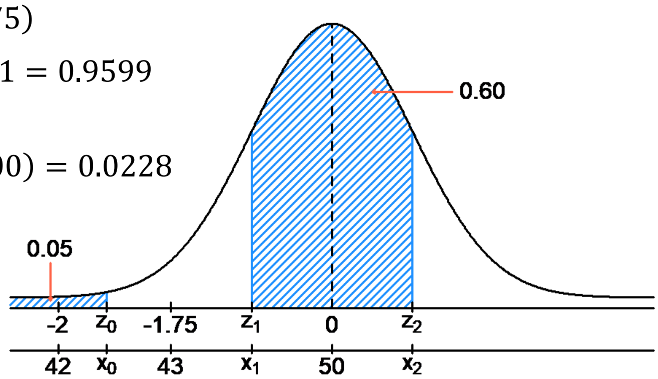
d) $1 - \mathrm{P}(Z < z_1) = 0.1587$
$\mathrm{P}(Z < z_1) = 1 - 0.1587 = 0.8413$
$\Rightarrow z_1 = 1.00$

Probleem 3 / *Problem 3*

$$X \sim N(\mu = 50;\ \sigma^2 = 4^2) \qquad \Rightarrow \qquad Z = \frac{X - 50}{4}$$

a) $\mathrm{P}(X > 43) = \mathrm{P}\left(Z > \dfrac{43 - 50}{4}\right) = \mathrm{P}(Z > -1.75)$
$= 1 - \mathrm{P}(Z < -1.75) = 1 - 0.0401 = 0.9599$

b) $\mathrm{P}(X < 42) = \mathrm{P}\left(Z < \dfrac{42 - 50}{4}\right) = \mathrm{P}(Z < -2.00) = 0.0228$



Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.

c) $P(Z < z_1) = 0.05 \quad \Rightarrow \quad z_1 = \dfrac{-1.64 + (-1.65)}{2} = -1.645$

$\Rightarrow -1.645 = \dfrac{X - 50}{4} \quad \Rightarrow \quad X = 4 \times (-1.645) + 50 = 43.42$

d) $$P(z_1 < Z < 0) + P(0 < Z < z_2) = 0.60$$

$P(z_1 < Z < 0) = \dfrac{0.60}{2} = 0.30 \quad$ *and* $\quad P(0 < Z < z_2) = \dfrac{0.60}{2} = 0.30$

Geval 1 / *Case 1*:

$P(z_1 < Z < 0) = 0.30$
$P(Z < 0) - P(Z < z_1) = 0.30$
$0.50 - P(Z < z_1) = 0.30$
$P(Z < z_1) = 0.50 - 0.30 = 0.20$
$\therefore z_1 = -0.84$

Dus, / Therefore,

$$-0.84 = \frac{x_1 - 50}{4}$$
$$\Longrightarrow x_1 = 4 \times (-0.84) + 50 = 46.64$$

Geval 2 / *Case 2*:

$P(0 < Z < z_2) = 0.30$
$P(Z < z_2) - P(Z < 0) = 0.30$
$P(Z < z_2) - 0.50 = 0.30$
$P(Z < z_2) = 0.50 + 0.30 = 0.80$
$\therefore z_2 = 0.84$

Dus, / Therefore,

$$0.84 = \frac{x_2 - 50}{4}$$
$$\Longrightarrow x_2 = 4 \times 0.84 + 50 = 53.36$$

$$\therefore 46.64 < X < 53.36$$

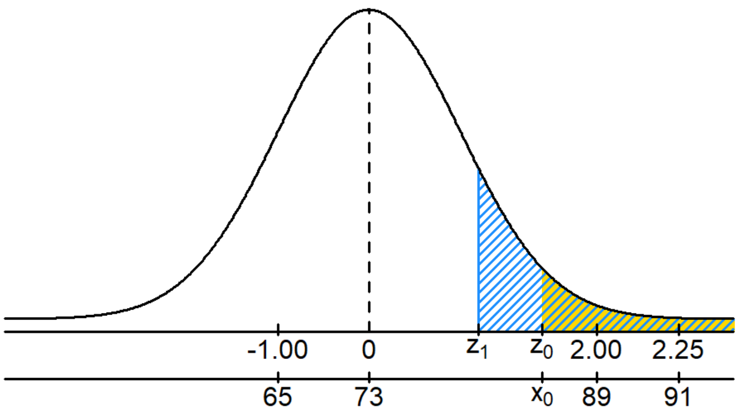
Probleem 4 / *Problem 4*

$$X \sim N(\mu = 73,\ \sigma^2 = 8^2)$$

a) $P(X < 91) = P\left(Z < \dfrac{91 - 73}{8}\right) = P(Z < 2.25) = 0.9878$

b) $P(65 < X < 89) = P\left(\dfrac{65 - 73}{8} < Z < \dfrac{89 - 73}{8}\right) = P(-1 < Z < 2)$
$= P(Z < 2) - P(Z < -1) = 0.9772 - 0.1587 = 0.8185$

c)
$P(Z > z_0) = 0.05$
$1 - P(Z < z_0) = 0.05$
$P(Z < z_0) = 1 - 0.05 = 0.95$
$z_0 = \dfrac{1.64 + 1.65}{2} = 1.645$
$1.645 = \dfrac{x_0 - 73}{8}$
$x_0 = 8 \times 1.645 + 73 = 86.160$



d) $P(Z > z_1) = 0.10 \quad \Longrightarrow \quad 1 - P(Z < z_1) = 0.10$
$P(Z < z_1) = 1 - 0.10 = 0.90 \quad \Longrightarrow \quad z_1 = 1.28$

Inleidende Statistiek / *Introductory Statistics*:

$$\frac{81 - 73}{8} = 1 < 1.28$$

Ander Kursus / *Other Course*:

$$\frac{68 - 62}{3} = 2 > 1.28$$

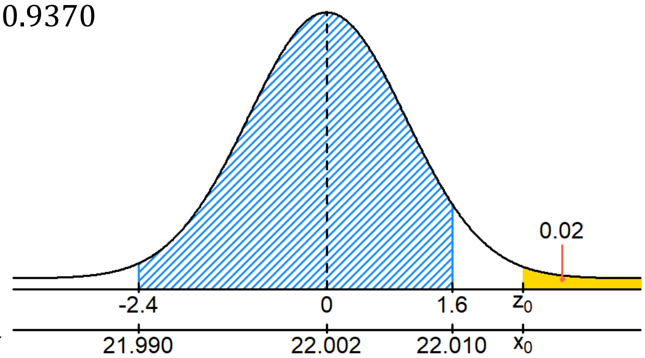
Met ’n punt van 68% in die ander kursus verdien ’n student ’n onderskeiding, terwyl ’n punt van 81% nie ’n onderskeiding in die inleidende statistiek-kursus behaal nie. Daarom is ’n student beter af met ’n 68% punt in die ander kursus. /
A mark of 68% *in the other course will earn a student a distinction while a mark of* 81% *will not earn a distinction in the introductory statistics course. Therefore, a student is better off with a* 68% *score in the other course.*

Probleem 5 / *Problem 5*

$$X \sim N(\mu = 22.002,\ \sigma^2 = 0.005^2) \qquad \Longrightarrow \qquad Z = \frac{X - 22.002}{0.005}$$

a) $$\mathrm{P}(21.99 < X < 22.01) = \mathrm{P}\left(\frac{21.990 - 22.002}{0.005} < Z < \frac{22.010 - 22.002}{0.005}\right)$$
$$= \mathrm{P}(-2.4 < Z < 1.6) = \mathrm{P}(Z < 1.6) - \mathrm{P}(Z < -2.4)$$
$$= 0.9452 - 0.0082 = 0.9370$$

b) $$\mathrm{P}(Z > z_0) = 0.02$$
$$1 - \mathrm{P}(Z < z_0) = 0.02$$
$$\mathrm{P}(Z < z_0) = 1 - 0.02 = 0.98$$
$$z_0 = 2.05$$
$$2.05 = \frac{x_0 - 22.002}{0.005}$$
$$x_0 = 2.05 \times 0.005 + 22.002 = 22.0123mm$$



c) $$X \sim N(\mu = 22.002,\ \sigma^2 = 0.004^2) \qquad \Longrightarrow \qquad Z = \frac{X - 22.002}{0.004}$$

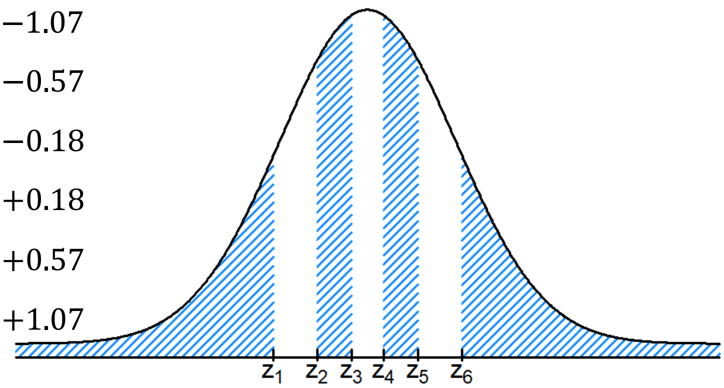
$$\mathrm{P}(21.99 < X < 22.01) = \mathrm{P}\left(\frac{21.990 - 22.002}{0.004} < Z < \frac{22.010 - 22.002}{0.004}\right)$$
$$= \mathrm{P}(-3 < Z < 2) = \mathrm{P}(Z < 2) - \mathrm{P}(Z < -3)$$
$$= 0.97720 - 0.00135 \approx 0.9759$$

$$2.05 = \frac{x_0 - 22.002}{0.004}$$
$$x_0 = 2.05 \times 0.004 + 22.002 = 22.0102mm$$

Probleem 6 / *Problem 6*

Die totale oppervlakte onder die normale digtheidskurwe = 1. Dit wil sê, $\mathrm{P}(-\infty < Z < \infty) = 1$. Om ses eweredig verdeelde waardes te kry, moet die totale digtheid dus in sewe (= 6 + 1) gelyke dele verdeel word. Sodat,/
The total area under the normal density curve $= 1$. *That is,* $\mathrm{P}(-\infty < Z < \infty) = 1$. *Therefore, to get six evenly distributed values requires dividing the total density into seven (*$= 6 + 1$*) equal slices. So that,*

$$\mathrm{P}(Z < z_1) = {}^{1}/_{7} = 0.1429 \quad \Rightarrow \quad z_1 = -1.07$$
$$\mathrm{P}(Z < z_2) = {}^{2}/_{7} = 0.2857 \quad \Rightarrow \quad z_2 = -0.57$$
$$\mathrm{P}(Z < z_3) = {}^{3}/_{7} = 0.4286 \quad \Rightarrow \quad z_3 = -0.18$$
$$\mathrm{P}(Z < z_4) = {}^{4}/_{7} = 0.5714 \quad \Rightarrow \quad z_4 = +0.18$$
$$\mathrm{P}(Z < z_5) = {}^{5}/_{7} = 0.7143 \quad \Rightarrow \quad z_5 = +0.57$$
$$\mathrm{P}(Z < z_6) = {}^{6}/_{7} = 0.8571 \quad \Rightarrow \quad z_6 = +1.07$$

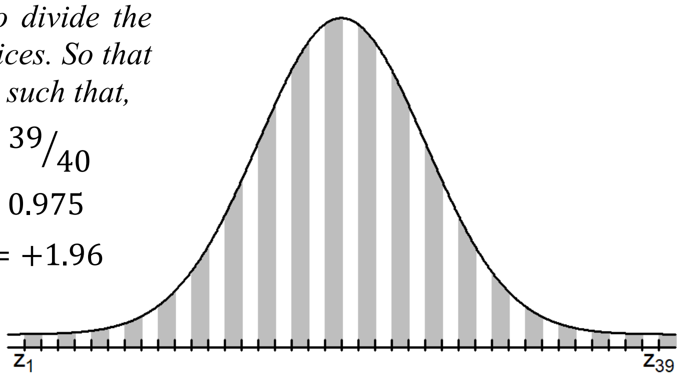


Probleem 7 / *Problem 7*

Die totale oppervlakte onder die normale digtheidskurwe = 1. Dit wil sê, $\mathrm{P}(-\infty < Z < \infty) = 1$. Om 39 waardes te vind wat eweredig versprei is, moet jy die totale digtheid in veertig (= 39 + 1) gelyke dele verdeel. Sodat die twee ekstreme waardes (z_1 en z_{39}) sodanig sal wees dat/

The total area under the normal density curve = 1. *That is,* $\mathrm{P}(-\infty < Z < \infty) = 1$. *Therefore, for there to be* 39 *evenly spread values, one needs to divide the total density into forty* (= 39 + 1) *equal slices. So that the two extreme values* (z_1 *and* z_{39}) *will be such that,*

$$\mathrm{P}(Z < z_1) = {}^{1}/_{40} \quad \blacksquare \quad \mathrm{P}(Z < z_{39}) = {}^{39}/_{40}$$
$$\mathrm{P}(Z < z_1) = 0.025 \quad \blacksquare \quad \mathrm{P}(Z < z_{39}) = 0.975$$
$$z_1 = -1.96 \quad \blacksquare \quad z_{39} = +1.96$$

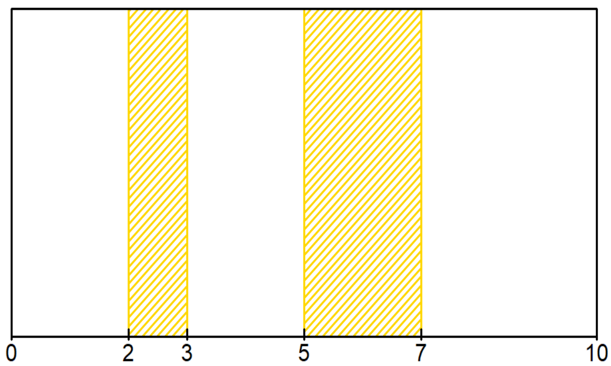


Probleem 8 / *Problem 8*

$$X \sim Uniform(a = 0,\ b = 10) \quad \Rightarrow \quad \mathrm{P}(X < t) = \frac{t-a}{b-a} = \frac{t-0}{10-0}; \quad 0 \le t \le 10$$

a) $\mathrm{P}(5 < X < 7) = \mathrm{P}(X < 7) - \mathrm{P}(X < 5)$
$$= \frac{7-0}{10-0} - \frac{5-0}{10-0} = \frac{2}{10} = 0.20$$

b) $\mathrm{P}(2 < X < 3) = \mathrm{P}(X < 3) - \mathrm{P}(X < 2)$
$$= \frac{3-0}{10-0} - \frac{2-0}{10-0} = \frac{1}{10} = 0.10$$



c) $\mu = \dfrac{a+b}{2} = \dfrac{0+10}{2} = 5$

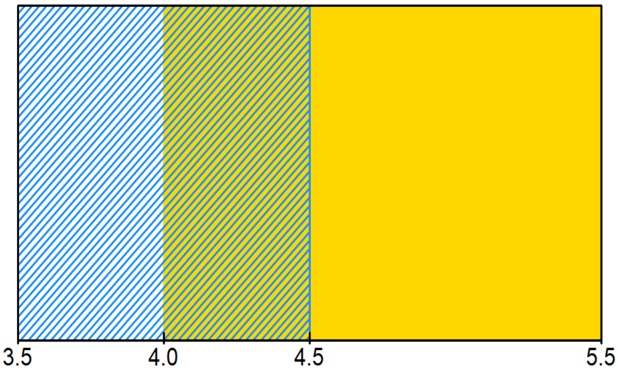
d) $\sigma = \sqrt{\dfrac{(b-a)^2}{12}} = \sqrt{\dfrac{(10-0)^2}{12}} \approx 2.8868$

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.

Probleem 9 / *Problem 9*

$$X \sim Uniform(a = 3.5, b = 5.5) \quad \Longrightarrow \quad \mathrm{P}(X < t) = \frac{t-a}{b-a} = \frac{t-3.5}{5.5-3.5}; \quad 3.5 \le t \le 5.5$$

a) $\mathrm{P}(X < 4.5) = \dfrac{4.5-3.5}{5.5-3.5} = \dfrac{1}{2} = 0.50$



b) $\mathrm{P}(X > 4.0) = 1 - \mathrm{P}(X < 4.0)$

$$= 1 - \frac{4.0-3.5}{5.5-3.5} = 1 - 0.25 = 0.75$$

c) $\mathrm{P}(4.0 < X < 4.5) = \mathrm{P}(X < 4.5) - \mathrm{P}(X < 4.0)$

$$= 0.50 - 0.25 = 0.25$$

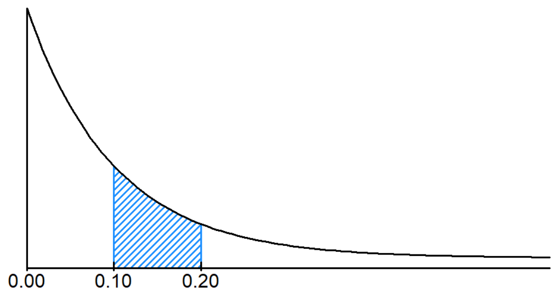
d) $\mu = \dfrac{a+b}{2} = \dfrac{3.5+5.5}{2} = 4.50 \text{ mins}$

$$\sigma = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{(5.5-3.5)^2}{12}} \approx 0.5774 \text{ mins} = 34.64 \text{ secs}$$

Probleem 10 / *Problem 10*

$$X \sim Exp(\lambda = 10) \quad \Longrightarrow \quad \mathrm{P}(X < t) = 1 - e^{-\lambda t} = 1 - e^{-10t}; \quad t \ge 0$$

a) $\mathrm{P}(X < 0.10) = 1 - e^{-10\times 0.10} \approx 0.6321$



b) $\mathrm{P}(X > 0.10) = 1 - \mathrm{P}(X < 0.10) \approx 1 - 0.6321 = 0.3679$

c) $\mathrm{P}(0.10 < X < 0.20) = \mathrm{P}(X < 0.20) - \mathrm{P}(X < 0.10)$

$$= 1 - e^{-10\times 0.20} - \mathrm{P}(X < 0.10)$$

$$\approx 0.8647 - 0.6321 = 0.2326$$

d) $\mathrm{P}(X < 0.10 \; or \; X > 0.20) = \mathrm{P}(X < 0.10) + \mathrm{P}(X > 0.20) - \mathrm{P}(X < 0.10 \; and \; X > 0.20)$

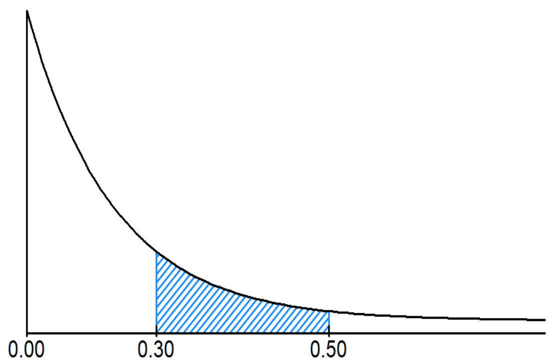
$$= \mathrm{P}(X < 0.10) + \mathrm{P}(X > 0.20) - 0 = \mathrm{P}(X < 0.10) + 1 - \mathrm{P}(X < 0.20)$$

$$\approx 0.6321 + 1 - 0.8647 = 0.7674$$

Probleem 11 / *Problem 11*

$$X \sim Exp(\lambda = 5) \quad \Longrightarrow \quad \mathrm{P}(X < t) = 1 - e^{-\lambda t} = 1 - e^{-5t}; \quad t \ge 0$$

a) $\mathrm{P}(X < 0.30) = 1 - e^{-5\times 0.30} \approx 0.7769$



b) $\mathrm{P}(X > 0.30) = 1 - \mathrm{P}(X < 0.30) \approx 1 - 0.7769 = 0.2231$

c) $\mathrm{P}(X > 0.30 \,\&\, X < 0.50) = \mathrm{P}(0.30 < X < 0.50)$

$$= \mathrm{P}(X < 0.50) - \mathrm{P}(X < 0.30)$$

$$= 1 - e^{-5\times 0.50} - \mathrm{P}(X < 0.30)$$

$$\approx 0.9179 - 0.7769 = 0.1410$$

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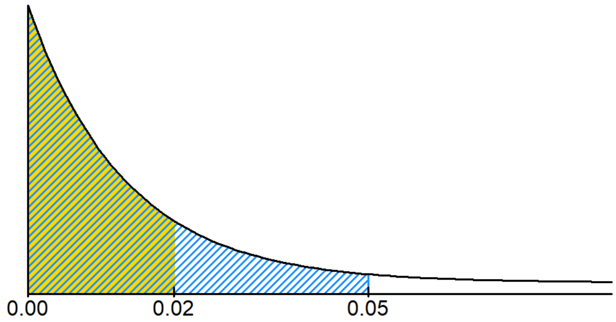
d) $P(X < 0.30 \text{ or } X > 0.50) = P(X < 0.30) + P(X > 0.50) - P(X < 0.30 \text{ and } X > 0.50)$

$= P(X < 0.30) + P(X > 0.50) - 0 = P(X < 0.30) + 1 - P(X < 0.50)$

$\approx 0.7769 + 1 - 0.9179 = 0.8590$

Probleem 12 / *Problem 12*

$$X \sim Exp(\lambda = 50/min) \quad \Longrightarrow \quad P(X < t) = 1 - e^{-\lambda t} = 1 - e^{-50t}; \quad t \geq 0$$



a) $P(X \leq 0.05) = 1 - e^{-50(0.05)} \approx 0.9179$

b) $P\left(X \leq \frac{1}{60}\right) = 1 - e^{-50(1/60)} \approx 0.5654$

c) $X \sim Exp(\lambda = 60)$

$P(X \leq 0.05) = 1 - e^{-60(0.05)} \approx 0.9502$

$P\left(X \leq \frac{1}{60}\right) = 1 - e^{-60\left(\frac{1}{60}\right)} \approx 0.6321$

Waarskynlikhede verhoog / *Probabilities increase*.

d) $X \sim Exp(\lambda = 30)$

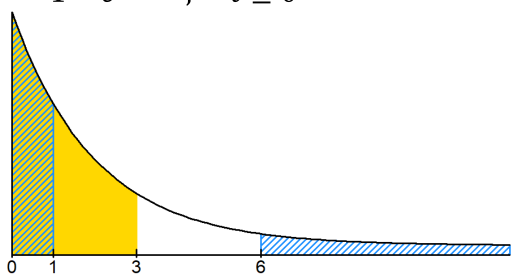
$P(X \leq 0.05) = 1 - e^{-30(0.05)} \approx 0.7769$

$P\left(X \leq \frac{1}{60}\right) = 1 - e^{-30\left(\frac{1}{60}\right)} \approx 0.3935$

Waarskynlikhede verminder / *Probabilities decrease*.

Probleem 13 / *Problem 13*

$$X \sim Exp\left(\lambda = \frac{12}{60} = 0.20/min\right) \quad \Longrightarrow \quad P(X < t) = 1 - e^{-\lambda t} = 1 - e^{-0.20t}; \quad t \geq 0$$



a) $P(X < 1) = 1 - e^{-0.20(1)} \approx 0.1813$

b) $P(X < 3) = 1 - e^{-0.20(3)} \approx 0.4512$

c) $P(X > 6) = 1 - P(X < 6) = 1 - \left(1 - e^{-0.20(6)}\right) \approx 0.3012$

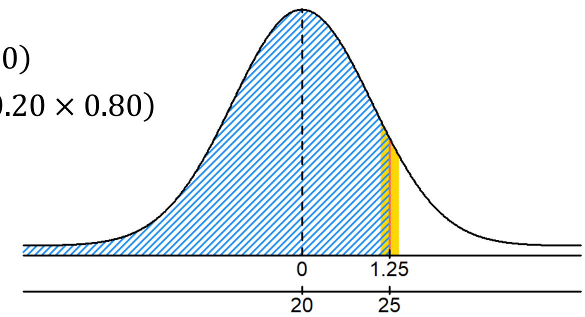
Probleem 14 / *Problem 14*

Laat $X_a = X$ benaderd / *Let* $X_a = X$ *approximate*.

$$X \sim Binomial(n, \pi) = Binomial(100, 0.20)$$

$$X_a \sim N(n\pi; n\pi[1-\pi]) = N(100 \times 0.20; \; 100 \times 0.20 \times 0.80)$$

$$X_a \sim N(20; \; 4^2) \quad \Longrightarrow \quad Z = \frac{X_a - 20}{4}$$



a) $\mathrm{P}(X = 25) \approx \mathrm{P}(24.5 < X_a < 25.5) = \mathrm{P}(X_a < 25.5) - \mathrm{P}(X_a < 24.5)$

$$= \mathrm{P}\left(Z < \frac{25.5 - 20}{4}\right) - \mathrm{P}\left(Z < \frac{24.5 - 20}{4}\right)$$

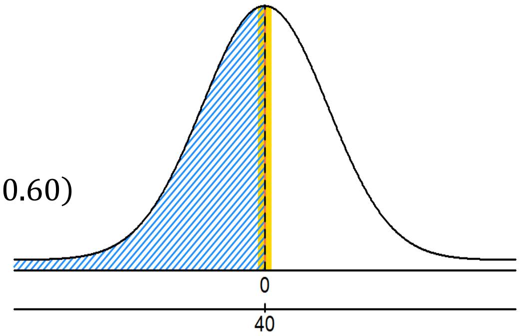
$$= \mathrm{P}(Z < 1.38) - \mathrm{P}(Z < 1.13) = 0.9162 - 0.8708 = 0.0454$$

b) $\mathrm{P}(X < 25) = \mathrm{P}(X \leq 24) \approx \mathrm{P}(X_a < 24.5) = 0.8708$

c) $\mathrm{P}(X \leq 25) \approx \mathrm{P}(X_a < 25.5) = \mathrm{P}(Z < 1.38) = 0.9162$

d) $\mathrm{P}(X > 25) = 1 - \mathrm{P}(X \leq 25) \approx 1 - \mathrm{P}(X_a < 25.5) \approx 1 - 0.9162 = 0.0838$

Probleem 15 / *Problem 15*



Laat $X_a = X$ benaderd / *Let* $X_a = X$ *approximate.*

$$X \sim Binomial(n, \pi) = Binomial(100, 0.40)$$

$$X_a \sim N(n\pi; n\pi[1 - \pi]) = N(100 \times 0.40;\ \ 100 \times 0.40 \times 0.60)$$

$$X_a \sim N(40; 24) \quad \Longrightarrow \quad Z = \frac{X_a - 40}{\sqrt{24}}$$

a) $\mathrm{P}(X = 40) \approx \mathrm{P}(39.5 < X_a < 40.5) = \mathrm{P}(X_a < 40.5) - \mathrm{P}(X_a < 39.5)$

$$= \mathrm{P}\left(Z < \frac{40.5 - 40}{\sqrt{24}}\right) - \mathrm{P}\left(Z < \frac{39.5 - 40}{\sqrt{24}}\right)$$

$$= \mathrm{P}(Z < 0.10) - \mathrm{P}(Z < -0.10) = 0.5398 - 0.4602 = 0.0796$$

b) $\mathrm{P}(X < 40) = \mathrm{P}(X \leq 39) \approx \mathrm{P}(X_a < 39.5) = \mathrm{P}(Z < -0.10) = 0.4602$

c) $\mathrm{P}(X \leq 40) \approx \mathrm{P}(X_a < 40.5) = \mathrm{P}(Z < 0.10) = 0.5398$

d) $\mathrm{P}(X > 40) = 1 - \mathrm{P}(X \leq 40) \approx 1 - \mathrm{P}(X_a < 40.5) \approx 1 - 0.5398 = 0.4602$

Probleem 16 / *Problem 16*

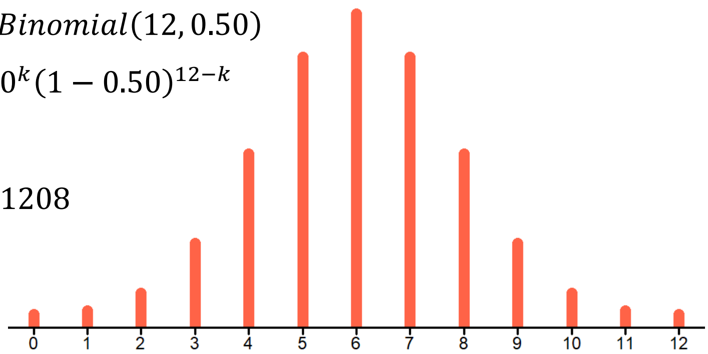
Laat X = gebeurtenis "head" word waargeneem. / *Let* X = *event head is observed.*
Laat $X_a = X$ benaderd / *Let* $X_a = X$ *approximate.*

<u>Eksakte Binomiaal Benadering / *Exact Binomial Estimation*</u>:

$$X \sim Binomial(n, \pi) = Binomial(12, 0.50)$$

$$\mathrm{P}(X = k) = \binom{12}{k} 0.50^k (1 - 0.50)^{12-k}$$

a) $\mathrm{P}(X = 4) = \binom{12}{4} 0.50^4 (1 - 0.50)^{12-4} \approx 0.1208$



b) $P(X \geq 4) = 1 - P(X \leq 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)]$

$$= 1 - \binom{12}{0} 0.50^{12} - \binom{12}{1} 0.50^{12} - \binom{12}{2} 0.50^{12} - \binom{12}{3} 0.50^{12}$$

$$\approx 1 - 0.0002 - 0.0029 - 0.0161 - 0.0537 = 0.9271$$

c) $P(5 \leq X \leq 7) = P(X = 5) + P(X = 6) + P(X = 7)$

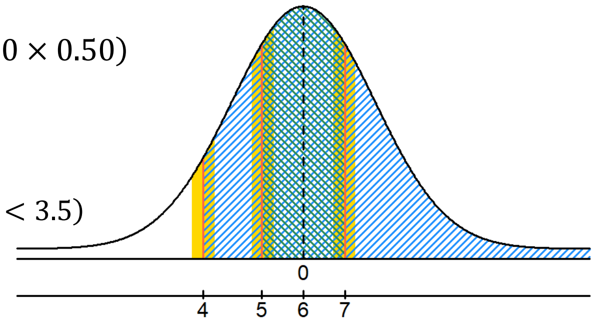
$$= \binom{12}{5} 0.50^5 \times 0.50^{12-5} + \binom{12}{6} 0.50^6 \times 0.50^{12-6} + \binom{12}{7} 0.50^7 \times 0.50^{12-7}$$

$$\approx 0.1934 + 0.2256 + 0.1934 = 0.6124$$

<u>Normaal Benadering/ *Normal Approximation*</u>:

$$X_a \sim N(n\pi; n\pi[1-\pi]) = N(12 \times 0.50;\ \ 12 \times 0.50 \times 0.50)$$

$$X_a \sim N(6; 3) \quad \Longrightarrow \quad Z = \frac{X_a - 6}{\sqrt{3}}$$



a) $P(X = 4) \approx P(3.5 < X_a < 4.5) = P(X_a < 4.5) - P(X_a < 3.5)$

$$= P\left(Z < \frac{4.5 - 6}{\sqrt{3}}\right) - P\left(Z < \frac{3.5 - 6}{\sqrt{3}}\right)$$

$$\approx P(Z < -0.87) - P(Z < -1.44) = 0.1922 - 0.0749 = 0.1173$$

b) $P(X \geq 4) = 1 - P(X \leq 3) \approx 1 - P(X_a < 3.5) \approx 1 - 0.0749 = 0.9251$

c) $P(5 \leq X \leq 7) \approx P(4.5 < X_a < 7.5) = P(X_a < 7.5) - P(X_a < 4.5)$

$$= P\left(Z < \frac{7.5 - 6}{\sqrt{3}}\right) - P\left(Z < \frac{4.5 - 6}{\sqrt{3}}\right)$$

$$\approx P(Z < 0.87) - P(Z < -0.87) = 0.8078 - 0.1922 = 0.6156$$

Probleem 17 / *Problem 17*

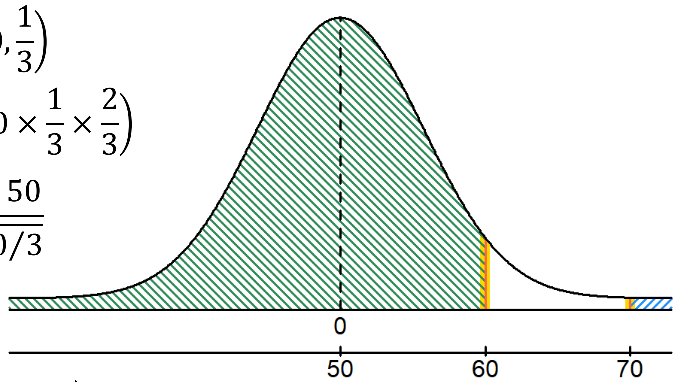
Laat X = gebeurtenis passasiers sal pretzels vir nagereg kies. /
Let X = event passenger chooses pretzels for dessert.

Laat $X_a = X$ benader / *Let $X_a = X$ approximate.*

$$X \sim Binomial(n, \pi) = Binomial\left(150, \frac{1}{3}\right)$$

$$X_a \sim N(n\pi; n\pi[1-\pi]) = N\left(150 \times \frac{1}{3};\ \ 150 \times \frac{1}{3} \times \frac{2}{3}\right)$$

$$X_a \sim N\left(50; \frac{100}{3}\right) \quad \Longrightarrow \quad Z = \frac{X_a - 50}{\sqrt{100/3}}$$



a) $P(X = 60) \approx P(59.5 < X_a < 60.5)$

$$= P(X_a < 60.5) - P(X_a < 59.5)$$

$$= P\left(Z < \frac{60.5 - 50}{\sqrt{100/3}}\right) - P\left(Z < \frac{59.5 - 50}{\sqrt{100/3}}\right)$$

$$\approx P(Z < 1.82) - P(Z < 1.65) = 0.9656 - 0.9505 = 0.0151$$

b) $P(X \geq 60) = 1 - P(X \leq 59) \approx 1 - P(X_a < 59.5) \approx 1 - 0.9505 = 0.0495$

c) $P(X < 60) = P(X \leq 59) \approx P(X_a < 59.5) \approx P(Z < 1.65) = 0.9505$

d) $P(X > 70) = 1 - P(X \leq 70) \approx 1 - P(X_a < 70.5)$

$$= 1 - P\left(Z < \frac{70.5 - 50}{\sqrt{100/3}}\right) \approx 1 - P(Z < 3.55)$$

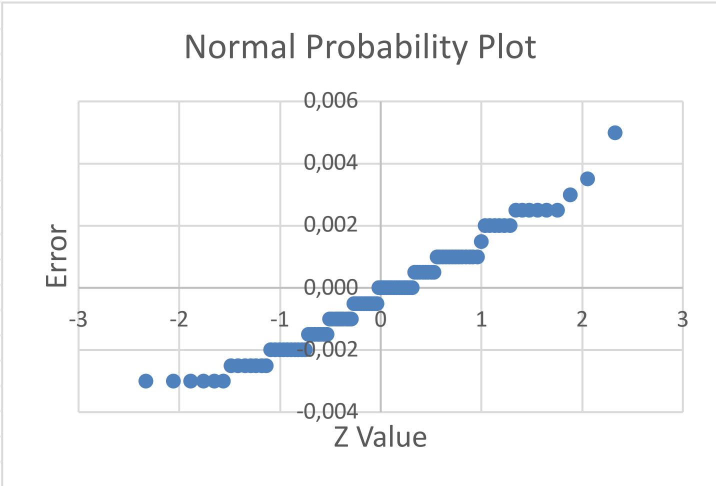
$$= 1 - 0.99981 = 0.00019$$

Probleem 18 / *Problem 18*

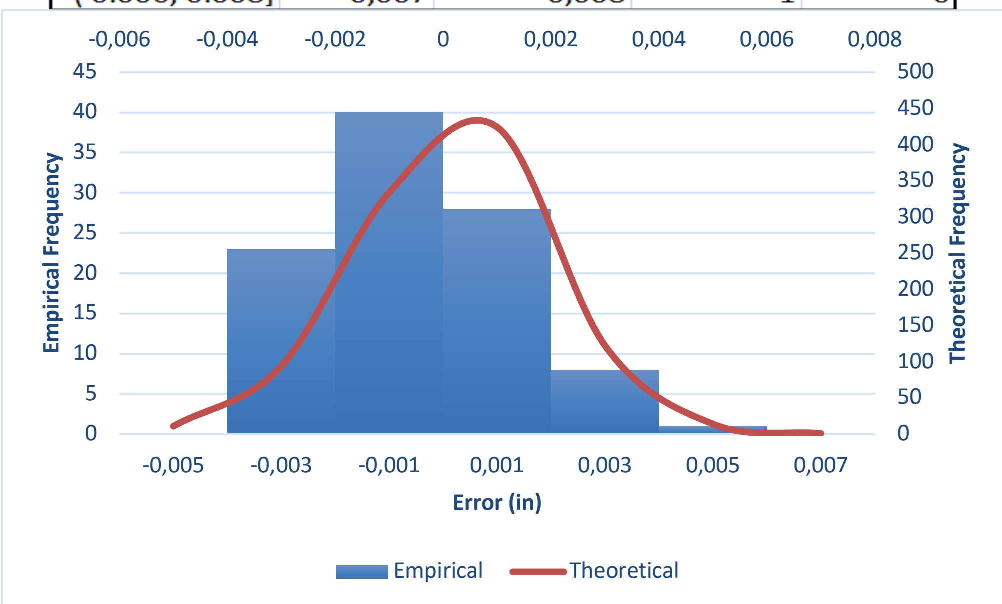
Mean:	Median:	Minimum:	Maximum:
-0.0002	0.0000	-0.0030	0.0050
Std. Dev.:	Skewness:	Kurtosis:	Count:
0.0017	0.3517	-0.1968	100
Q1 Position:	Q1 Value:	Q3 Position:	Q3 Value:
25.0000	-0.0015	76.0000	0.0010
			Range:
			0.0080

Empirical Rule

		Count	Proportion
$\mu - \sigma$	$\mu + \sigma$		
-0.0019	0.0015	61	0.61
$\mu - 2\sigma$	$\mu + 2\sigma$		
-0.0036	0.0032	98	0.98
$\mu - 3\sigma$	$\mu + 3\sigma$		
-0.0053	0.0049	99	0.99



Interval	Mid-point	Upper Bound	Theoretical	Empirical
(-0.006,-0.004]	-0,005	-0,004	11	0
(-0.004,-0.002]	-0,003	-0,002	94	23
(-0.002,-0.000]	-0,001	0,000	334	40
(0.000, 0.002]	0,001	0,002	424	28
(0.002, 0.004]	0,003	0,004	122	8
(0.004, 0.006]	0,005	0,006	14	1
(0.006, 0.008]	0,007	0,008	1	0



Characteristic	Observation	Implication
`Mean & Median`	Approximately equal to zero	Approximately symmetric
`Data proportion within` $\mu \pm \sigma$	Empirical 61% (Theoretical 68%)	Approximately symmetric
`Data proportion within` $\mu \pm 2\sigma$	Empirical 98% (Theoretical 95%)	Approximately symmetric
`Data proportion within` $\mu \pm 3\sigma$	Empirical 99% (Theoretical 99.7%)	Approximately symmetric
`Skewness`	Empirical 0.35 (Theoretical 0.00)	Slight rightward skewness
`Kurtosis`	Empirical -0.20 (Theoretical 0.00)	Slightly platykurtic
`Max – Q2 & Q2 – Min`	Approximately equal	Approximately symmetric
`Q3 – Q2 & Q2 – Q1`	Approximately equal	Approximately symmetric
`Max – Q3 & Q1 – Min`	Approximately equal	Approximately symmetric

<u>Conclusion</u>: Staalmetingsfout is ongeveer normaal verdeel. Hierdie afleiding word ondersteun deur die lineêre patroon wat blyk uit die normaal waarskynlikheidsgrafiek sowel as die benaderde ooreenkoms tussen die empiriese en die teoretiese digthede op die histogram./ *Steel measurement error is approximately normally distributed. This inference is supported by the linear pattern evident from the normal probability plot as well as the approximate agreement between the empirical and the theoretical densities on the normal histogram plot.*

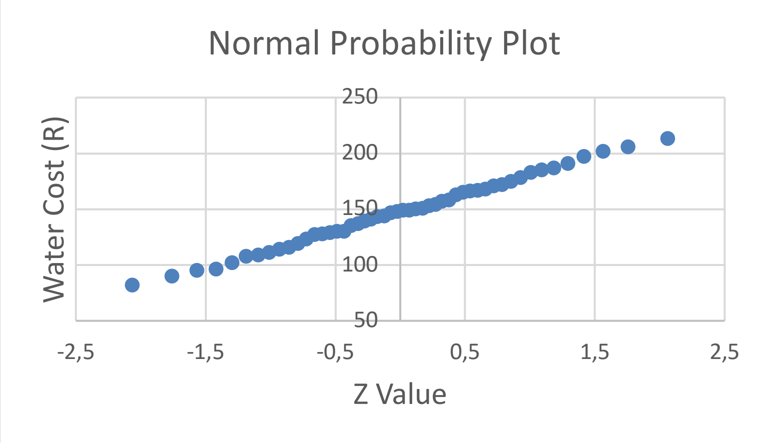
Probleem 19 / *Problem 19*

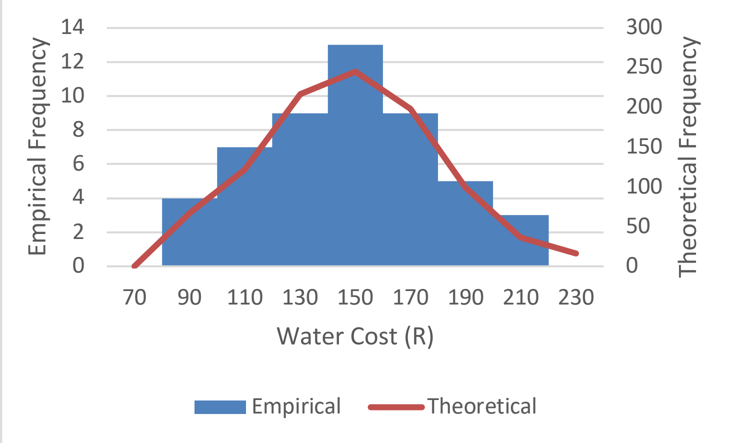
Mean:	Median:	Minimum:	Maximum:
147.0600	148.5000	82	213
Std. Dev.:	Skewness:	Kurtosis:	Count:
31.6914	0.0158	-0.5442	50
Q1 Position:	Q1 Value:	Q3 Position:	Q3 Value:
13	127	38	168
			Range:
			131

Empirical Rule			
		Count	Proportion
$\mu - \sigma$	$\mu + \sigma$		
115.3686	178.7514	33	0.66
$\mu - 2\sigma$	$\mu + 2\sigma$		
83.6772	210.4428	48	0.96
$\mu - 3\sigma$	$\mu + 3\sigma$		
51.9859	242.1341	50	1.00

Interval	Mid-point	Upper Bound	Empirical	Theoretical
(60; 80]	70	80	0	0
(80; 100]	90	100	4	67
(100; 120]	110	120	7	122
(120; 140]	130	140	9	217
(140; 160]	150	160	13	245
(160; 180]	170	180	9	198
(180; 200]	190	200	5	99
(200; 220]	210	220	3	36
(220; 240]	230	240	0	16

Levine, D. M., Stephan, D. F. & Szabat K. A.. (2021). *Statistics for Managers Using Microsoft Excel*. 9th edition, Pearson.





Characteristic	Observation	Implication
Mean & Median	Approximately equal	Approximately symmetric
Data proportion within $\mu \pm \sigma$	Empirical 66% (Theoretical 68%)	Approximately symmetric
Data proportion within $\mu \pm 2\sigma$	Empirical 96% (Theoretical 95%)	Approximately symmetric
Data proportion within $\mu \pm 3\sigma$	Empirical 100% (Theoretical 99.7%)	Approximately symmetric
Skewness	Empirical 0.02 (Theoretical 0.00)	Approximately symmetric
Kurtosis	Empirical -0.54 (Theoretical 0.00)	Slightly platykurtic
Max – Q2 & Q2 – Min	Approximately equal	Approximately symmetric
Q3 – Q2 & Q2 – Q1	Approximately equal	Approximately symmetric
Max – Q3 & Q1 – Min	Equal	Symmetric

Note: For the purpose for this course:

a) neem aan dat veranderlikes met skeefheid en kurtose waardes binne die interval (-0.5; 0.5) ongeveer normaal verdeel is. /
assume that variable with skewness and kurtosis values within the interval (-0.5; 0.5) are approximately normally distributed.

Conclusion: Waterkoste data is ongeveer normaal verdeel. Hierdie afleiding word ondersteun deur die lineêre patroon wat blyk uit die normaal waarskynlikheidsgrafiek sowel as die benaderde ooreenkoms tussen die empiriese en die teoretiese digthede op die histogram./ *Water cost data is approximately normally distributed. This inference is supported by the linear pattern evident from the normal probability plot as well as the approximate agreement between the empirical and the theoretical densities on the normal histogram plot.*

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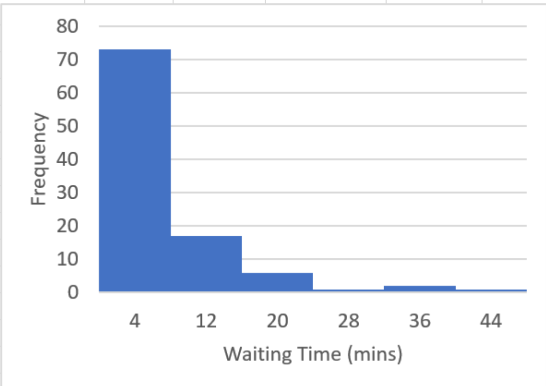
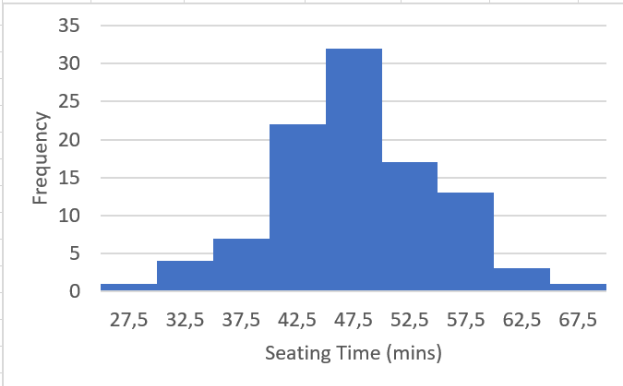
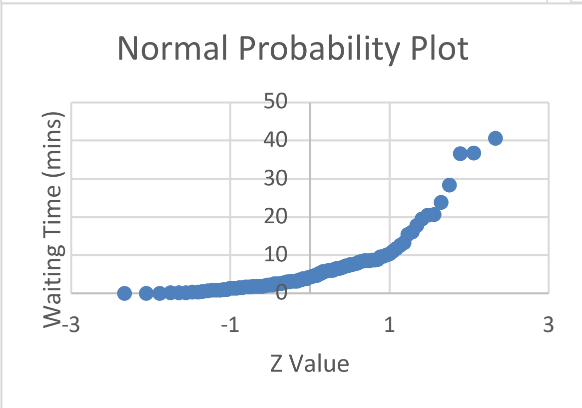
Probleem 20 / *Problem 20*

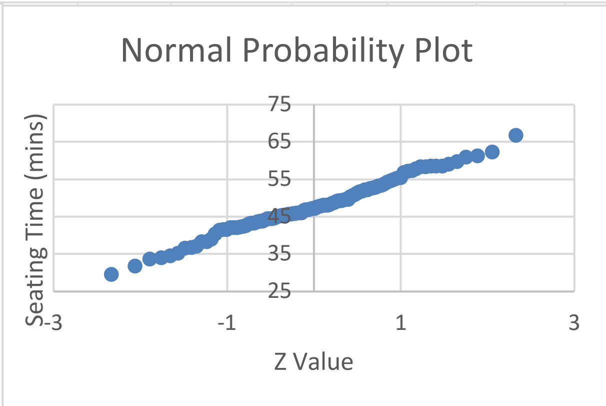
Waiting Time (mins)			
Interval	**Mid-point**	**Upper Bound**	**Frequency**
[0; 8]	4	8	73
(8; 16]	12	16	17
(16; 24]	20	24	6
(24; 32]	28	32	1
(32; 40]	36	40	2
(40; 48]	44	48	1

Seating Time (mins)			
Interval	**Mid-point**	**Upper Bound**	**Frequency**
(25; 30]	27,5	30	1
(30; 35]	32,5	35	4
(35; 40]	37,5	40	7
(40; 45]	42,5	45	22
(45; 50]	47,5	50	32
(50; 55]	52,5	55	17
(55; 60]	57,5	60	13
(60; 65]	62,5	65	3
(65; 70]	67,5	70	1









a) Wagtyd lyk soos 'n eksponensiaal verdeling omdat dit skeef na regs is. /
Waiting time resembles an exponential distribution because it is skewed to the right.

b) Sit tyd lyk soos 'n normale verdeling omdat dit ongeveer simmetries is. /
Seating time resembles a normal distribution because it is approximately symmetric.